

Vijaya Sai Dasari

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Summary

AI/ML Engineer with 4+ years of experience designing, developing, and deploying scalable artificial intelligence solutions across enterprise environments. Skilled in machine learning, deep learning, generative AI, large language models, NLP, predictive analytics, and MLOps. Experienced in building production-ready AI systems, optimizing algorithms, and implementing cloud-based architectures using modern frameworks and automation practices to deliver reliable, scalable, and impactful intelligent solutions.

Certificates

- Python for Data Science and Machine Learning Bootcamp | AWS Fundamentals: Going Cloud-Native**

Technical Skills

- Programming & Data Engineering:** Python, SQL, Bash Scripting, Pandas, NumPy, Apache Spark, PySpark, Hive, Presto, BigQuery, Data Processing, Data Transformation, ETL Pipelines, Feature Engineering, Distributed Computing, Real-Time Data Processing, Data Streaming, Kafka, AWS Kinesis
- Machine Learning, Deep Learning & AI:** Supervised Learning, Unsupervised Learning, Predictive Modeling, Anomaly Detection, Classification, Regression, XGBoost, LightGBM, Random Forest, Isolation Forest, Autoencoders, Time-Series Forecasting, PyTorch, TensorFlow, Keras, Scikit-learn, Transformers, BERT, Reinforcement Learning, DQN, PPO, Ray RLlib, Model Optimization, Hyperparameter Tuning
- Generative AI, NLP & Knowledge Systems:** Generative AI, Large Language Models (LLMs), Hugging Face Transformers, OpenAI API, LangChain, vLLM, Ollama, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Agentic Workflows, QLoRA Fine-Tuning, Natural Language Processing, Semantic Search, Vector Databases, FAISS, Pinecone, ChromaDB, Weaviate, Knowledge Graph Integration, Embeddings
- MLOps, Deployment & Cloud Platforms:** MLflow, Kubeflow, Docker, Kubernetes, Amazon EKS, Azure Kubernetes Service (AKS), FastAPI, gRPC, Apache Airflow, CI/CD Pipelines, Model Monitoring, Model Drift Detection, Model Governance, ONNX Deployment, AWS SageMaker, AWS Bedrock, AWS Lambda, Amazon S3, Azure AI Studio, Azure Data Factory, Google Vertex AI, Cloud-Native AI Deployment
- AI Operations, Explainability & Analytics:** Explainable AI (XAI), SHAP, LIME, Captum, AI Governance, Responsible AI, Model Interpretability, Production AI Systems, Observability, Logging, Prometheus, Grafana, Streamlit, Plotly Dash, Real-Time Dashboards, Performance Monitoring, Technical Documentation, Agile Development

Professional Experience

AI/ML Engineer, Fisher Investments

08/2024 – 03/2026 | Remote, USA

- Designed and deployed an AI-driven Market Anomaly Detection Platform using PyTorch, TensorFlow, Apache Spark, agentic AI architectures, and MCP integrations with LLM-based reasoning pipelines, improving enterprise portfolio risk signal identification accuracy by 12%.
- Built real-time financial intelligence pipelines using Kafka, AWS Lambda, Elasticsearch, RAG workflows, multimodal AI agents, and autonomous market monitoring systems, reducing analyst workload through advanced predictive insights and scalable decision support frameworks globally.
- Developed reinforcement learning solutions for portfolio rebalancing using Ray RLlib, SageMaker, and agentic AI frameworks, enabling adaptive strategy testing across market scenarios with scalable simulations and increasing expected returns by 5% through optimization.
- Applied Explainable AI and agentic AI methods using SHAP, LIME, Captum, MCP, and tool-using AI agents to deliver clear recommendations, strengthening compliance, governance, auditability, and stakeholder trust across critical enterprise portfolio decision workflows.
- Established end-to-end MLOps workflows with Kubeflow, MLflow, Docker, EKS, and CI/CD automation, with documentation and cross-functional collaboration for training, validation, deployment, monitoring & governance of production AI models, reducing operational incidents by 10%.
- Created interactive AI dashboards and alerting systems using Plotly Dash, FastAPI, Prometheus, and generative AI copilots with MCP-enabled orchestration, delivering real-time portfolio insights, risk metrics, predictive analytics, and faster advisor decision-making across enterprises.

ML Engineer, Verizon India Pvt Ltd

08/2020 – 07/2023 | Chennai, India

- Architected an AI-powered telecom analytics framework using anomaly detection, graph-based learning, and ensemble algorithms to uncover complex network degradation patterns, minimize false alarms & boost predictive incident response across distributed infrastructure settings.
- Constructed and optimized large-scale data engineering workflows using Azure Data Factory, PySpark, SQL, and Delta Lake to ingest, transform, and analyze billions of network events, CDRs, and telemetry streams for real-time intelligence generation.
- Engineered production-grade ML pipelines utilizing XGBoost, LightGBM, Autoencoders, and Isolation Forest models to predict network failures, detect fraudulent activities, and analyze subscriber behavior, delivering 92% F1-score and improving automated operational decisions.
- Implemented advanced NLP-driven solutions with BERT embeddings, semantic analysis, and classification techniques on customer interactions and support records, extracting actionable insights, automating issue routing, reducing analyst workload, and improving service prioritization.
- Operationalized end-to-end ML deployment workflows through Docker, Azure Kubernetes Service, MLflow, Airflow, Prometheus, and Grafana, enabling continuous integration, automated retraining, model observability, drift monitoring, and scalable production inference systems.

ML Intern, Verizon India Pvt Ltd

01/2020 – 07/2020 | Chennai, India

- Assisted in developing machine learning models for telecom datasets by supporting data preprocessing, feature engineering, exploratory analysis, and model testing using Python and Scikit-learn, gaining practical experience in predictive analytics and ML workflows.
- Supported the team in preparing datasets, creating visualizations, and evaluating model performance through SQL and Python-based analysis, contributing to research activities focused on improving network insights and understanding customer behavior patterns.

Education

Illinois Institute of Technology, Chicago, IL, USA – Master of Science in Computer Science

08/2023 – 05/2025

RMD Engineering College, Chennai, India – Bachelor of Engineering in Computer Science

08/2019 – 05/2023

Projects

MEDCOMPARE – AI Powered Medication Data Benchmarking Tool

- Built a real-time Streamlit tool benchmarking medication data from LLMs using semantic similarity, fuzzy matching, and biostatistics; aligned outputs with FDA and RxNorm standards; supported batch evaluation, scoring, and FHIR exports for clinical review.

Classification with XGBoost + ONNX Deployment

- Developed an XGBoost classifier on 1.2M entries, outperforming baselines; applied scaling, outlier handling, and data prep; converted model to ONNX for real-time edge/cloud deployment, showcasing scalable predictive analytics in IoT and embedded environments.